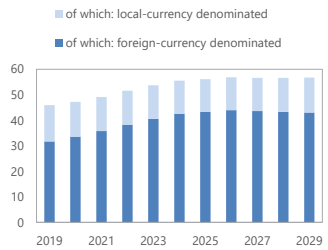
Table 2. Madagascar: Public Sector Debt Sustainability Framework, Baseline Scenario, 2016-2039

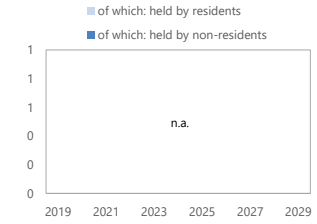
(In percent of GDP; unless otherwise indicated)

	Actual			Projections								Average 6/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
Public sector debt 1/	**47.1**	**46.0**	**45.7**	**46.0**	**47.2**	**49.2**	**51.6**	**53.7**	**55.6**	**56.8**	**56.1**	**42.3**	**53.3**
of which: external debt	**29.6**	**29.5**	**30.5**	**31.8**	**33.6**	**35.8**	**38.3**	**40.6**	**42.6**	**43.0**	**37.8**	**28.7**	**40.0**
Change in public sector debt	-3.5	-1.1	-0.3	0.3	1.3	2.0	2.4	2.1	1.8	0.1	-0.4		
Identified debt-creating flows	**-3.9**	**-3.8**	**-2.4**	**-0.9**	**1.1**	**1.9**	**2.5**	**2.4**	**2.1**	**0.1**	**-0.4**	**-0.2**	**1.0**
Primary deficit	**0.4**	**1.6**	**0.6**	**1.3**	**3.5**	**4.3**	**4.7**	**4.7**	**4.5**	**2.4**	**1.5**	**1.6**	**3.4**
Revenue and grants	14.6	14.8	14.8	16.4	15.0	14.8	14.4	14.6	14.6	15.2	15.7	**12.7**	**15.0**
of which: grants	3.4	2.9	2.8	3.9	2.4	1.9	1.4	1.2	0.9	0.7	0.6		
Primary (noninterest) expenditure	15.0	16.4	15.5	17.7	18.5	19.1	19.2	19.3	19.1	17.6	17.2	**14.2**	**18.4**
Automatic debt dynamics	**-4.3**	**-5.4**	**-3.0**	**-2.9**	**-3.0**	**-2.9**	**-2.8**	**-2.8**	**-2.9**	**-2.8**	**-2.4**		
Contribution from interest rate/growth differential	-3.6	-3.2	-3.7	-2.9	-2.8	-2.6	-2.4	-2.5	-2.6	-2.5	-2.2		
of which: contribution from average real interest rate	-1.5	-1.3	-1.4	-0.7	-0.4	-0.3	-0.2	-0.2	-0.2	-0.1	0.2		
of which: contribution from real GDP growth	-2.0	-1.9	-2.3	-2.3	-2.3	-2.3	-2.3	-2.4	-2.5	-2.5	-2.3		
Contribution from real exchange rate depreciation	-0.7	-2.2	0.7	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.6**	**0.6**	**0.6**	**0.5**	**0.5**	**0.5**	**0.5**	**0.5**	**0.0**	**0.5**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.6	0.6	0.6	0.5	0.5	0.5	0.5	0.5		
Residual	**0.4**	**2.7**	**2.1**	**1.2**	**-0.1**	**-0.3**	**-0.4**	**-0.5**	**-0.6**	**-0.3**	**-0.3**	**2.0**	**-0.2**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**34.0**	**34.1**	**34.9**	**36.3**	**38.0**	**39.5**	**40.9**	**43.6**	**45.8**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**229.5**	**208.2**	**232.7**	**244.7**	**263.2**	**270.7**	**280.2**	**287.2**	**292.0**		
Debt service-to-revenue and grants ratio 3/	**7.0**	**7.0**	**43.5**	**39.4**	**45.6**	**51.6**	**56.2**	**59.9**	**62.0**	**70.0**	**96.7**		
Gross financing need 4/	1.5	2.7	7.1	8.9	11.1	12.6	13.4	14.0	14.1	13.6	17.2		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	4.2	4.3	5.2	5.2	5.3	5.1	4.9	4.8	4.8	4.6	4.3	**2.3**	**4.9**
Average nominal interest rate on external debt (in percent)	0.8	0.9	1.1	1.1	1.4	1.4	1.4	1.5	1.5	1.6	1.6	**0.7**	**1.5**
Average real interest rate on domestic debt (in percent)	-7.8	-6.1	-6.8	-3.7	-1.6	-0.8	0.1	0.2	0.1	1.0	1.9	**-6.7**	**-0.2**
Real exchange rate depreciation (in percent, + indicates depreciation)	-2.5	-7.7	2.4	...	...	...	...	...	...	...	...	**1.1**	**...**
Inflation rate (GDP deflator, in percent)	8.5	6.5	7.3	6.7	6.3	5.8	5.2	5.0	5.0	5.0	5.0	**7.2**	**5.3**
Growth of real primary spending (deflated by GDP deflator, in percent)	8.5	14.0	-0.7	20.5	10.3	8.4	5.2	5.4	4.0	4.7	3.3	**1.9**	**6.3**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	3.9	2.7	0.9	1.1	2.3	2.3	2.3	2.5	2.7	2.4	1.9	**2.5**	**2.4**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, central bank, government-guaranteed debt, non-guaranteed SOE debt . Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

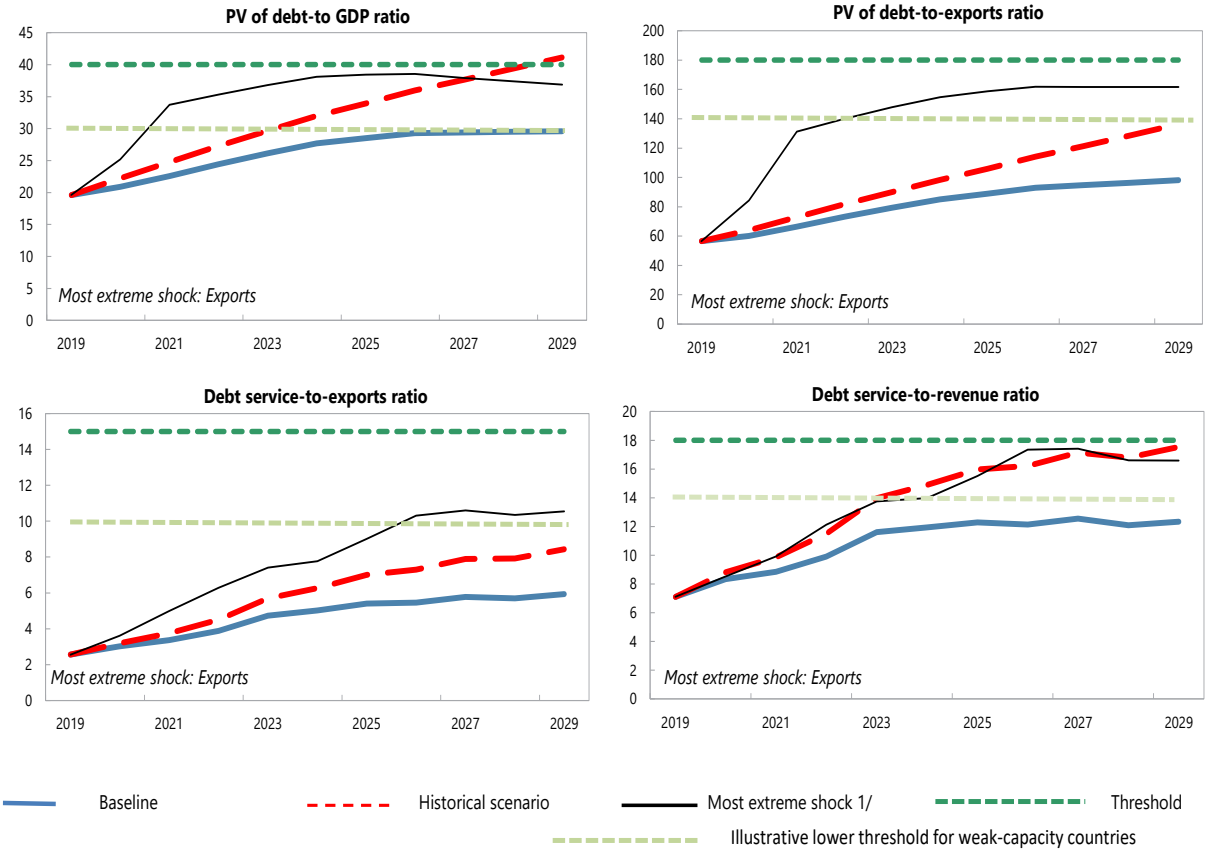
4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

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Figure 1. Madagascar: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2019-2029



Customization of Default Settings		
	Size	Interactions
Tailored Stress		
Combined CL	No	
Natural disaster	No	No
Commodity price	Yes	Yes
Market financing	n.a.	n.a.

Borrowing assumptions on additional financing needs resulting from the stress tests*		
	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.6%	1.6%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	26	26
Avg. grace period	4	4

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

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